

DBMS – PRODUCT AND CATEGORY MANAGEMENT

Project Title: EcoCart – Sustainable Products E-Commerce Order Management System

Skill Task: DBMS

Week: 2

1. Database Design

1.1 Category Table

Attribute	Data Type	Constraint
<i>Category_ID</i>	INT	Primary Key
<i>Category_Name</i>	VARCHAR(100)	NOT NULL, UNIQUE
<i>Description</i>	VARCHAR(200)	—

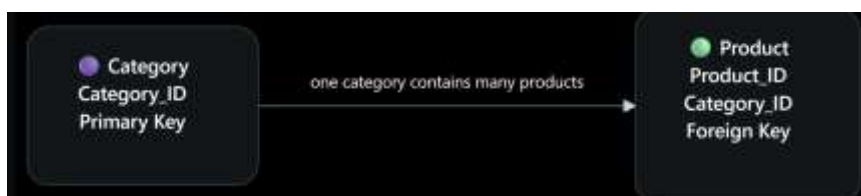
1.2 Product Table

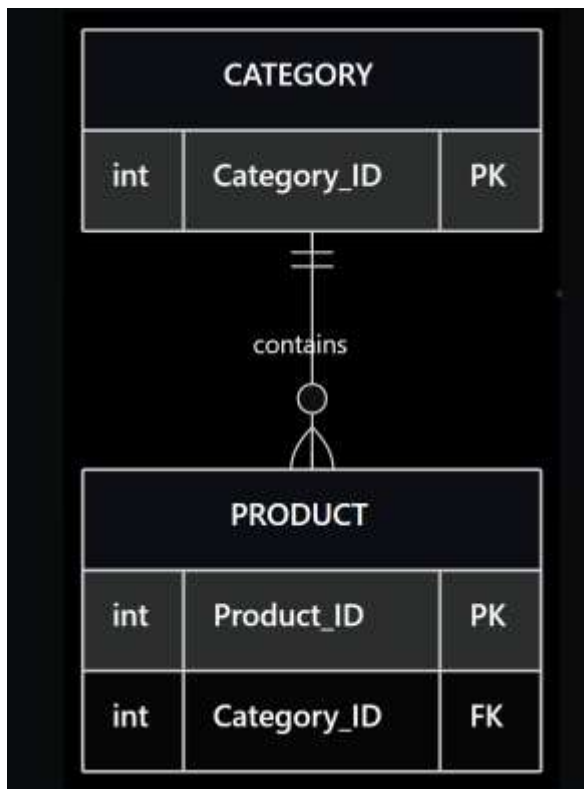
Attribute	Data Type	Constraint
<i>Product_ID</i>	INT	Primary Key
<i>Product_Name</i>	VARCHAR(100)	NOT NULL
<i>Category_ID</i>	INT	Foreign Key
<i>Price</i>	NUMBER(10,2)	NOT NULL
<i>Stock_Quantity</i>	INT	NOT NULL

2. Primary and Foreign Key Relationship

Category → Product

- Category.Category_ID → Primary Key
- Product.Product_ID → Primary Key
- Product.Category_ID → Foreign Key referencing Category.Category_ID
- Relationship: **One-to-Many (1:M)**





3. Table Creation

3.1 Create Category Table

```

SQL> CREATE TABLE Category (
2     Category_ID INT PRIMARY KEY,
3     Category_Name VARCHAR(100) NOT NULL UNIQUE,
4     Description VARCHAR(200)
5 );

Table created.
  
```

3.2 Create Product Table

```

SQL> CREATE TABLE Product (
2     Product_ID INT PRIMARY KEY,
3     Product_Name VARCHAR(100) NOT NULL,
4     Category_ID INT NOT NULL,
5     Price NUMBER(10,2) NOT NULL,
6     Stock_Quantity INT NOT NULL,
7     FOREIGN KEY (Category_ID) REFERENCES Category(Category_ID)
8 );

Table created.
  
```

4. Data Insertion

4.1 Category Records

```
SQL> INSERT INTO Category
  2 VALUES (1, 'Spices', 'Natural and organic spices');

1 row created.

SQL> INSERT INTO Category
  2 VALUES (2, 'Organic Food', 'Organic food products');

1 row created.

SQL> INSERT INTO Category
  2 VALUES (3, 'Eco Friendly Home', 'Environment friendly household products');

1 row created.

SQL> INSERT INTO Category
  2 VALUES (4, 'Natural Products', 'Natural and sustainable products');

1 row created.

SQL> INSERT INTO Category
  2 VALUES (5, 'Eco Friendly Personal Care', 'Natural personal care products');
```

4.2 Product Records

```
SQL> INSERT INTO Product
  2 VALUES (101, 'Organic Turmeric Powder', 1, 120.00, 50);

1 row created.

SQL> INSERT INTO Product
  2 VALUES (102, 'Black Pepper', 1, 180.00, 40);

1 row created.

SQL> INSERT INTO Product
  2 VALUES (103, 'Organic Honey', 2, 350.00, 25);

1 row created.

SQL> INSERT INTO Product
  2 VALUES (104, 'Bamboo Storage Box', 3, 450.00, 15);

1 row created.

SQL>
SQL> INSERT INTO Product
  2 VALUES (105, 'Natural Herbal Soap', 4, 100.00, 60);

1 row created.

SQL> COMMIT;
```



5. Data Retrieval

5.1 Display Category Records

```
SQL> SELECT * FROM Category;

CATEGORY_ID
CATEGORY_NAME
DESCRIPTION
-----
1
Spices
Natural and organic spices

2
Organic Food
Organic food products

CATEGORY_ID
CATEGORY_NAME
DESCRIPTION
-----

3
Eco Friendly Home
Environment friendly household products

4
Natural Products

CATEGORY_ID
CATEGORY_NAME
DESCRIPTION
-----
Natural and sustainable products

5
Eco Friendly Personal Care
```

5.2 Display Product Records

```
SQL> SELECT * FROM Product;
```

```
PRODUCT_ID
```

```
PRODUCT_NAME
```

```
CATEGORY_ID    PRICE STOCK_QUANTITY
```

```
101
Organic Turmeric Powder
1          120          50
```

```
102
Black Pepper
1          180          40
```

```
PRODUCT_ID
```

```
PRODUCT_NAME
```

```
CATEGORY_ID    PRICE STOCK_QUANTITY
```

```
103
Organic Honey
2          350          25
```

```
104
Bamboo Storage Box
```

```
PRODUCT_ID
```

```
PRODUCT_NAME
```

```
CATEGORY_ID    PRICE STOCK_QUANTITY
```

```
3          450          15
```

```
105
Natural Herbal Soap
```

6. Product Update

Update the price and stock quantity of Organic Turmeric Powder.

```
SQL> UPDATE Product
2 SET Price = 130.00,
3     Stock_Quantity = 45
4 WHERE Product_ID = 101;
```

```
1 row updated.
```

7. Product Deletion

Delete the product Natural Herbal Soap.

```
SQL> DELETE FROM Product
      2 WHERE Product_ID = 105;

1 row deleted.
```

8. Category-wise Product Reports

8.1 Product Listing by Category

This report displays the category name along with the products belonging to each category.

```
SQL> SELECT
      2     c.Category_Name,
      3     p.Product_Name,
      4     p.Price,
      5     p.Stock_Quantity
      6 FROM Category c
      7 JOIN Product p
      8 ON c.Category_ID = p.Category_ID
      9 ORDER BY c.Category_Name;
```

CATEGORY_NAME

PRODUCT_NAME

PRICE STOCK_QUANTITY

Eco Friendly Home

Bamboo Storage Box

450

15

Organic Food

Organic Honey

350

25

CATEGORY_NAME

PRODUCT_NAME

PRICE STOCK_QUANTITY

Spices

Black Pepper

180

40

Spices

Organic Turmeric Powder

8.2 Products in Spices Category

```
SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     p.Price,
  5     p.Stock_Quantity
  6 FROM Product p
  7 JOIN Category c
  8 ON p.Category_ID = c.Category_ID
  9 WHERE c.Category_Name = 'Spices';
```

PRODUCT_ID

PRODUCT_NAME

PRICE STOCK_QUANTITY

PRODUCT_ID	PRODUCT_NAME	PRICE	STOCK_QUANTITY
101	Organic Turmeric Powder	130	45

PRODUCT_ID	PRODUCT_NAME	PRICE	STOCK_QUANTITY
102	Black Pepper	180	40

PRODUCT_ID

PRODUCT_NAME

PRICE STOCK_QUANTITY

8.3 Product Count by Category

```
SQL> SELECT
  2     c.Category_Name,
  3     COUNT(p.Product_ID) AS Total_Products
  4 FROM Category c
  5 LEFT JOIN Product p
  6 ON c.Category_ID = p.Category_ID
  7 GROUP BY c.Category_Name
  8 ORDER BY c.Category_Name;
```

CATEGORY_NAME

TOTAL_PRODUCTS

CATEGORY_NAME	TOTAL_PRODUCTS
Eco Friendly Home	1

CATEGORY_NAME	TOTAL_PRODUCTS
Eco Friendly Personal Care	0

CATEGORY_NAME	TOTAL_PRODUCTS
Natural Products	0

CATEGORY_NAME

TOTAL_PRODUCTS

CATEGORY_NAME	TOTAL_PRODUCTS
Organic Food	1

CATEGORY_NAME	TOTAL_PRODUCTS
Spices	2